

Disproof of Conjecture 00000009617: the even shift has a 2-state Fischer cover but Hankel rank 3

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2026-10-04

Abstract

Conjecture 00000009617 asserts, among other things, that the Fischer (“Fisher”) minimal state count of a sofic shift equals the rank of the Hankel matrix of its language. We show that this clause is false. For the even shift X (binary sequences in which every run of 1s between two 0s has even length) the Fischer cover has exactly 2 states, and no presentation of any kind has fewer. But the Hankel matrix $H[u][v] = [uv \in \mathcal{L}(X)]$ has the 3×3 minor $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ with determinant -1 , so its rank is at least 3 (in fact exactly 3) over every field. The decisive facts (the presented shift, the minimal state count 2, the language for all words, and the nonsingular minor) are formalized and checked in a self-contained Lean 4 project. An independent Python script confirms them.

1 The conjecture

The statement (English version) reads:

Definition: Sofic shifts are factor images of SFTs, presented by finite graphs with output (hidden Markov models); the Fisher minimal realization is unique. *Conjecture:* The Fisher minimal state count of a sofic shift equals the rank of the Hankel matrix of its language; the rank is computable in polynomial time by a local criterion on words of length $2n + 1$, and the minimal realization graph structure is unique.

The Chinese version says the same. The conjecture is a conjunction of three clauses. We refute the first one:

(C) *For every sofic shift X , the number of states of its Fischer (minimal right-resolving) presentation equals the rank of the Hankel matrix of $\mathcal{L}(X)$.*

A conjunction with a false clause is false, so the conjecture is false. The two later clauses are not needed and we say nothing about them. “Fisher” is a common misspelling of R. Fischer, who introduced the minimal right-resolving presentation [1].

2 Definitions

We follow Lind and Marcus [2]. The alphabet is $\{0, 1\}$. A point is a bi-infinite sequence $x = (x_i)_{i \in \mathbb{Z}}$. A word $w = w_0 \cdots w_{n-1}$ occurs in x at i if $x_{i+j} = w_j$ for all $j < n$. The language $\mathcal{L}(X)$ of a set X of points is the set of words that occur in some point of X .

Labelled graphs and presentations. A labelled graph G on the vertex set $\{0, \dots, n-1\}$ is a set of edges $p \xrightarrow{a} q$ with $a \in \{0, 1\}$. The shift X_G presented by G is the set of label sequences of bi-infinite paths, i.e. of all x for which there are vertices s_i ($i \in \mathbb{Z}$) with an edge $s_i \xrightarrow{x_i} s_{i+1}$ for every i . A shift space X is *sofic* if $X = X_G$ for some finite G ; then G *presents* X . (Parallel edges with the same label do not change X_G , so we may take G to be a set of labelled edges.) G is *right-resolving* if the out-edges of each vertex carry distinct labels.

Fischer cover. Every irreducible sofic shift has a minimal right-resolving presentation, unique up to isomorphism [1], [2, §3.3]. It is called the Fischer cover. It has the fewest vertices among all right-resolving presentations [2, §3.3]. So the “Fisher minimal state count” is

$$\text{Fis}(X) = \min\{n : X \text{ has a right-resolving presentation with } n \text{ vertices}\}.$$

We also consider $\text{Pres}(X) = \min\{n : X \text{ has a presentation with } n \text{ vertices}\}.$

Hankel matrix and rank. The Hankel matrix of a language $\mathcal{L}$ is the infinite 0/1 matrix indexed by pairs of words, $H[u][v] = 1$ if $uv \in \mathcal{L}$ and 0 otherwise. Its row u is the indicator of the follower set $\mathcal{F}(u) = \{v : uv \in \mathcal{L}\}$. The rank of an infinite matrix over a field K is the dimension of its row space. Equivalently, it is the supremum of the ranks of its finite submatrices, or the largest r such that some $r \times r$ minor is nonzero in K . This is the meaning of “Hankel rank” in realization theory, from Kronecker’s theorem on Hankel matrices of rational functions to the Fliess and Carlyle–Paz theorems on rational formal power series and the Hankel rank of hidden Markov processes [3, 4, 5].

3 The even shift and its language

The even shift is

$$X = \{x \in \{0, 1\}^{\mathbb{Z}} : \text{no block } 01^{2k+1}0 \text{ (} k \geq 0 \text{) occurs in } x\}.$$

A block $01^m 0$ is exactly a run of m ones bounded by 0s on both sides, so X consists of the sequences in which every such run has even length.

Lemma 1. *For every word w : $w \in \mathcal{L}(X)$ if and only if w contains no factor $01^{2k+1}0$.*

Proof. If w occurs in $x \in X$ at i , then a factor $01^{2k+1}0$ of w at position j would occur in x at $i+j$. Conversely, if w has no such factor, let x be w placed at positions $0, \dots, |w| - 1$ and 1 everywhere else. Every 0 of x lies inside w . So a block $01^{2k+1}0$ of x would lie inside w , which is impossible. Hence $x \in X$ and $w \in \mathcal{L}(X)$. $\square$

4 The Fischer cover has two states

Let G be the graph with vertices $A = 0$, $B = 1$ and edges

$$A \xrightarrow{0} A, \quad A \xrightarrow{1} B, \quad B \xrightarrow{1} A.$$

It is right-resolving and irreducible. It is follower-separated: 0 labels a path from A but not from B .

Proposition 2. $X_G = X$.

Proof. ($X_G \subseteq X$.) Let (s_i) be a path with labels (x_i) . Suppose $01^{2k+1}0$ occurs at i . The only 0-edge ends at A , so $s_{i+1} = A$. Along the 1-edges the state alternates, so $s_{i+1+t} = A$ for even t and B for odd t ($t \leq 2k+1$). Hence $s_{i+2k+2} = B$. But $x_{i+2k+2} = 0$ and B has no 0-edge. This is a contradiction.

($X \subseteq X_G$.) Let $x \in X$. We say position i is in state B if one of the following holds:

- (a) there is a 0 at or after i , and the first one is at $i+t$ with t odd;
- (b) there is no 0 at or after i , there is a 0 before i , and the last one is at $i-1-t$ with t odd;
- (c) x has no 0 at all and i is odd.

Otherwise i is in state A . We check that $s_i \xrightarrow{x_i} s_{i+1}$ is an edge for every i .

If $x_i = 0$, then (a) fails at i with $t = 0$, and (b), (c) fail too. So $s_i = A$. At $i+1$, (a) would give a block 01^t0 at i with t odd, which is excluded since $x \in X$. (b) fails because the last 0 before $i+1$ is at i , so $t = 0$. (c) fails. So $s_{i+1} = A$ and $A \xrightarrow{0} A$ is an edge.

If $x_i = 1$, we show that s_{i+1} is the other state from s_i . Then the step is $A \xrightarrow{1} B$ or $B \xrightarrow{1} A$. If there is a 0 after i , say first at $i+1+t$, then the first 0 at or after i is at $i+(t+1)$. Only rule (a) applies at both positions, with parities of t and $t+1$. If there is no 0 after i but one before i , say last at $i-1-t$, then the last 0 before $i+1$ is at $(i+1)-1-(t+1)$. Only rule (b) applies, with parities of t and $t+1$. If x has no 0 at all, rule (c) applies with the parities of i and $i+1$. $\square$

Proposition 3. *No labelled graph with at most one vertex presents X .*

Proof. A graph with no vertex has no bi-infinite path, but $0^\infty \in X$. A graph with one vertex and loop label set $T \subseteq \{0, 1\}$ presents the full shift $T^\mathbb{Z}$. If $0 \notin T$, it misses $0^\infty \in X$. If $1 \notin T$, it misses $1^\infty \in X$ (which has no 0). If $T = \{0, 1\}$, it contains $\cdots 0001000\cdots \notin X$, since this point contains the block 010. $\square$

Theorem 4. $\text{Fis}(X) = \text{Pres}(X) = 2$. *The graph G is the Fischer cover of X .*

Proof. By Propositions 2 and 3. The Fischer cover is the unique minimal right-resolving presentation, so it is G . $\square$

The verification script also enumerates all 2^8 labelled graphs on two vertices. Exactly two of them have the same blocks as X up to length 8, and these are G and its relabelling.

5 The Hankel matrix has rank 3

By Lemma 1, $H[u][v] = 1$ iff uv has no factor $01^{\text{odd}}0$. Take rows $u \in \{11, 0, 01\}$ and columns $v \in \{0, 10, 1\}$:

$$\begin{array}{c|ccc}
 & 0 & 10 & 1 \\
\hline
11 & 110 \in \mathcal{L} & 1110 \in \mathcal{L} & 111 \in \mathcal{L} \\
0 & 00 \in \mathcal{L} & 010 \notin \mathcal{L} & 01 \in \mathcal{L} \\
01 & 010 \notin \mathcal{L} & 0110 \in \mathcal{L} & 011 \in \mathcal{L}
\end{array}
\quad M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

We have $\det M = 1 \cdot (0 - 1) - 1 \cdot (1 - 0) + 1 \cdot (1 - 0) = -1$. The integer matrix $N = \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$ satisfies $MN = NM = I$. So M is invertible over every commutative ring, and in particular over every field.

Theorem 5. *Over every field K , the Hankel matrix of $\mathcal{L}(X)$ has rank exactly 3.*

Proof. The rank is at least 3 because M is a nonsingular submatrix. For the upper bound we show that H has at most four distinct rows, one of which is zero. If $u \notin \mathcal{L}(X)$, then $\mathcal{F}(u) = \emptyset$, because $\mathcal{L}(X)$ is closed under taking factors. Let $u \in \mathcal{L}(X)$. A forbidden block in uv lies inside u (impossible), inside v , or straddles the junction. A straddling block 01^m0 has its first 0 in u and all 1s up to the junction, so it starts at the last 0 of u . Write $u = u'01^j$ if u contains a 0, and $v = 1^r0v'$ if v contains a 0. Then $uv \in \mathcal{L}(X)$ iff $v \in \mathcal{L}(X)$ and, when both contain a 0, $j + r$ is even. Hence $\mathcal{F}(u) = \mathcal{L}(X)$ if u has no 0, $\mathcal{F}(u) = \mathcal{F}(0)$ if j is even, and $\mathcal{F}(u) = \mathcal{F}(01)$ if j is odd. The rows lie in the span of the three vectors $\mathbf{1}_{\mathcal{L}(X)}$, $\mathbf{1}_{\mathcal{F}(0)}$ and $\mathbf{1}_{\mathcal{F}(01)}$, so the rank is at most 3. $\square$

Corollary 6. $\text{Fis}(X) = 2 \neq 3 = \text{rank}_K H$ for every field K . Clause (C), and with it Conjecture 00000009617, is false.

6 Readings of the clause

- **Rank over a field** ($\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_p$, or any other field). The rank is $3 \neq 2$. The determinant -1 is a unit in every ring, so the counterexample does not depend on the field. This is the standard meaning of the rank of a Hankel matrix (Section 2).
- **Fisher count.** We read it as the size of the minimal right-resolving presentation. If it means the least number of states of *any* presentation, it is still 2 (Theorem 4). If it means the left Fischer cover (minimal left-resolving presentation), it is also 2: X is closed under reversal, and reversing G gives a left-resolving presentation; Proposition 3 holds for all graphs.
- **Indexing of H .** Rows and columns may range over all words, over nonempty words, or over words of length at most n for any $n \geq 2$. All these contain M , so the rank is at least 3 in each case. Indexing by u^Rv or transposing does not change the rank.
- **Boolean rank (disclosure).** Over the Boolean semiring (OR/AND), the rank of H is 2. Indeed $\mathbf{1}_{\mathcal{L}(X)} = \mathbf{1}_{\mathcal{F}(0)} \vee \mathbf{1}_{\mathcal{F}(01)}$: if v has no 0, then $v \in \mathcal{F}(0)$; otherwise $v \in \mathcal{F}(0)$ or $v \in \mathcal{F}(01)$ by the parity of its leading run of 1s. The fooling set $\{(0,0), (01,10)\}$ shows that the Boolean rank is at least 2. Under this reading the even shift gives $2 = 2$ and is *not* a counterexample. We do not claim anything about the Boolean reading. It is not the standard meaning: Boolean rank is a different notion and is always named as such. Over $\mathbb{Q}$ the vector $\mathbf{1}_{\mathcal{L}(X)}$ is not $\mathbf{1}_{\mathcal{F}(0)} + \mathbf{1}_{\mathcal{F}(01)}$, since both follower sets contain every word 1^k . This is why the field rank is 3.
- **Hidden Markov model reading.** One could instead take the Hankel matrix of the probability function $P(w)$ of a measure on X . That matrix is not determined by the shift or by “its language”. The clause speaks of the Hankel matrix of the language, so we use the indicator function of $\mathcal{L}(X)$.
- **Other covers.** The number of distinct nonempty follower sets of words of X is 3. This is the size of the follower-set graph and of the minimal deterministic automaton of $\mathcal{L}(X)$ without its dead state, and it equals the Hankel rank here. But that graph is not minimal, and it is not the Fischer cover. The Fischer cover only keeps the follower sets of synchronizing words, namely $\mathcal{F}(0)$ and $\mathcal{F}(01)$.

7 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project. It uses only the core library (no `Mathlib`). Points are functions $\text{Int} \rightarrow \text{Bool}$ (`false` is 0, `true` is 1).

- `EvenShift x`: no `ForbiddenAt x i k`, the block $01^{2k+1}0$ at i . `Lang X w`: some point of X contains w .
- `LGraph n`, `IsPathLabel`, `Presents E X` (X equals the set of bi-infinite path labels), `RightResolving`, `Sofic`. `IsFisherCount X m` means that m is the least number of vertices of a right-resolving presentation. `IsMinPresCount` is the same for arbitrary presentations.
- `hankelEntry X u v` is 1 if `Lang X (u ++ v)` holds and 0 otherwise. `minor`, and `det` (Laplace expansion). `HankelRankAtLeast X r` means that some $r \times r$ minor has nonzero integer determinant, i.e. the rank over $\mathbb{Q}$ is at least r . `HankelRankAtLeastMod p` is the same over $\mathbb{Z}/p$. `HankelRankIs X m` means at least m and not at least $m + 1$.
- `Claim`: for every sofic X and every m with `IsFisherCount X m`, `HankelRankIs X m`. `ClaimMod p` and `ClaimAnyPresentation` are the variants for $\mathbb{Z}/p$ and for arbitrary presentations.
- `lang_iff`: Lemma 1, for all words. `fisher_sound`, `fisher_complete`, `fisher_presents`: Proposition 2. `no_zero_state`, `no_one_state`: Proposition 3. `fisher_count`, `min_presentation_count`: Theorem 4. `fisher_rightResolving` and `fisher_irreducible_separated` check the properties of G .
- `minor_eq`, `minor_det` ($= -1$), `minor_unimodular` (the inverse N), `rows_independent` (the three rows are linearly independent over $\mathbb{Q}$), `hankel_rank_ge_three` and `hankel_rank_ge_three_mod` (for every $p \geq 2$).
- `conjecture_00000009617_false` : $\neg \text{Claim}$. Also `conjecture_00000009617_false_mod` ($\forall p \geq 2, \neg \text{ClaimMod } p$) and `conjecture_00000009617_false_any_presentation`. `fisher_lt_rank`: every m with `IsFisherCount X m` equals 2, and the rank is at least $m + 1$.

Non-vacuity: `evenShift_sofic` and `fisher_count` show that the hypotheses of `Claim` hold for X . `allZero_mem`, `allOne_mem` and `single_one_not_mem` show that X is neither empty nor full. The final examples show that `Lang` is nontrivial and that `det` returns 0 on a singular matrix.

Limitations. Lean proves rank *at least* 3, which suffices. The upper bound in Theorem 5 and the Boolean-rank remark are proved in this report and checked on finite truncations by `verify.py`, but not in Lean. Lean works over the alphabet $\{0, 1\}$. The clause quantifies over all sofic shifts, so a binary counterexample suffices. The project has no `sorry`, no `native_decide` and no added axioms. `#print axioms` reports only the standard `propext`, `Classical.choice` and `Quot.sound`. Classical logic is used to define the path in `fisher_complete` and the Hankel entries of an arbitrary shift.

8 Independent check

`verify.py` (Python 3 standard library) does the following:

- It checks that three descriptions of $\mathcal{L}(X)$ agree on all words of length at most 14: forbidden blocks, parities of inner runs, and paths in G .
- It enumerates all labelled graphs with 0, 1 or 2 vertices. None with fewer than 2 vertices matches, and the matching 2-vertex graphs are G up to relabelling.
- It recomputes the minor, its determinant -1 and its inverse.
- It computes the rank of the Hankel truncation on all 127 words of length at most 6: it is 3 over $\mathbb{Q}$ and over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$.
- It confirms that only 4 distinct rows occur.
- It confirms the Boolean-rank identity.

9 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex
```

References

- [1] R. Fischer, Sofic systems and graphs, *Monatsh. Math.* 80 (1975), 179–186.
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- [4] J. W. Carlyle and A. Paz, Realizations by stochastic finite automata, *J. Comput. System Sci.* 5 (1971), 26–40.
- [5] H. Ito, S.-I. Amari and K. Kobayashi, Identifiability of hidden Markov information sources and their minimum degrees of freedom, *IEEE Trans. Inform. Theory* 38 (1992), 324–333.